IPMH: Integration of stepped care for perinatal mood and anxiety disorders among women attending MCH clinics

Data and Safety Monitoring Board (DSMB): Interim Analysis

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Date of data freeze: 2026-03-30

Person-time calculations

50% expected person-time calculation

Total expected person-time was estimated as the mean time from enrollment to delivery (calculated as mean gestational age at delivery minus mean gestational age at enrollment), plus 14 weeks postpartum, multiplied by the total expected sample size of 2,970 participants, yielding a total expected person-time of 66826.98 weeks. 50% of the expected person-time is 33413.49 weeks.

Current total person-time elapsed to date calculation

As of the data freeze date of 2026-03-30, the total person-time elapsed among enrolled participants is 49685.57 weeks. This represents 74.35% of the expected person-time. Because the current person-time elapsed is more than 50% of the expected person-time, we proceed with the interim analysis as planned.

Interim analysis results

Generalized linear mixed models (GLMM) with a Gaussian family and identity link were used to compare change in PHQ-9 and GAD-7 scores from enrollment to 14 weeks postpartum between arms. Models included fixed effects for treatment arm, time since enrollment, and their interaction, and a random effect for facility.

PHQ-9

```r
phq_model <- lmer(phq9_score ~ arm_random * time + (1 | clt_study_site), data = analysis_df)

summary(phq_model)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: phq9_score ~ arm_random * time + (1 | clt_study_site)
##    Data: analysis_df
##
## REML criterion at convergence: 15477.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1538 -0.7005 -0.0775  0.5424  6.3873
##
## Random effects:
##  Groups         Name        Variance Std.Dev.
##  clt_study_site (Intercept) 2.077    1.441
##  Residual                   7.481    2.735
## Number of obs: 3173, groups:  clt_study_site, 20
##
## Fixed effects:
##                              Estimate Std. Error t value
## (Intercept)                  7.052257   0.463886  15.203
## arm_randomIntervention      -0.057300   0.655450  -0.087
## time                        -0.176107   0.006539 -26.932
## arm_randomIntervention:time  0.021073   0.009281   2.270
##
## Correlation of Fixed Effects:
##             (Intr) arm_rI time
## arm_rndmInt -0.708
## time        -0.105  0.074
## arm_rndmIn:  0.074 -0.102 -0.705
```

GAD-7

```r
gad_model <- lmer(gad7_score ~ arm_random * time + (1 | clt_study_site), data = analysis_df)

summary(gad_model)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: gad7_score ~ arm_random * time + (1 | clt_study_site)
##    Data: analysis_df
##
## REML criterion at convergence: 15578.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4695 -0.6943 -0.1228  0.5809  5.4283
##
## Random effects:
##  Groups         Name        Variance Std.Dev.
##  clt_study_site (Intercept) 2.036    1.427
##  Residual                   7.690    2.773
## Number of obs: 3176, groups:  clt_study_site, 20
##
## Fixed effects:
##                              Estimate Std. Error t value
## (Intercept)                  6.227800   0.459638  13.549
## arm_randomIntervention       0.078579   0.649415   0.121
## time                        -0.154417   0.006635 -23.275
## arm_randomIntervention:time  0.019649   0.009406   2.089
##
## Correlation of Fixed Effects:
##             (Intr) arm_rI time
## arm_rndmInt -0.708
## time        -0.107  0.076
## arm_rndmIn:  0.075 -0.104 -0.705
```